

NULL Values & 3-Valued Logic

Q. Explain the concept of NULL in SQL. How do we check for NULL values, handle them in expressions, and how do they affect comparisons and aggregate functions?

📌 Definition to Remember

NULL in SQL is a special marker that represents **missing, unknown, or inapplicable data**. It is NOT the same as zero (0) or an empty string (""). NULL literally means "no value." SQL uses **Three-Valued Logic** (True, False, **Unknown**) when dealing with NULL in comparisons.

1. Key Rules About NULL

- **NULL ≠ 0** (zero is a number; NULL is the absence of a value)
- **NULL ≠ ""** (empty string is a valid text value)
- Any arithmetic involving NULL → result is **NULL** (e.g., `1000 + NULL = NULL`)
- Any comparison with NULL → result is **UNKNOWN** (e.g., `Age > NULL = UNKNOWN`)

2. Checking for NULL Values

You **cannot** use `= NULL` or `!= NULL` — these always return UNKNOWN. Use specialized operators:

```
-- IS NULL: find rows where a column has no value
SELECT * FROM Employee WHERE Manager_ID IS NULL;

-- IS NOT NULL: find rows where a column has a value
SELECT * FROM Employee WHERE Phone IS NOT NULL;
```

3. Three-Valued Logic

SQL uses True, False, and **Unknown** (instead of just True/False):

Condition	Result
<code>Salary = NULL</code>	Unknown (not False!)
<code>NULL = NULL</code>	Unknown (not True!)
<code>IS NULL</code> check	Correct way — returns True/False

4. Handling NULL with COALESCE()

`COALESCE(val1, val2, ...)` returns the **first non-NULL value** in the list — used to substitute a default value for NULLs.

```
-- Treat NULL commission as 0 when calculating total pay
SELECT Salary + COALESCE(Commission, 0) AS Total_Pay FROM Employee;
```

5. Effect on Aggregate Functions

- `SUM()` , `AVG()` , `MAX()` , `MIN()` → **automatically ignore NULL rows.**
- `COUNT(*)` → counts **all rows** including NULLs.
- `COUNT(column)` → **excludes NULL** rows.

Example: Salaries = [100, 200, NULL] → `AVG(Salary)` = $(100+200)/2 = 150$ (NULL ignored)

★ Must-Write Points (for 10 marks)

1. NULL represents **missing or unknown data** — NOT zero or empty string.
2. You **cannot** use `= NULL` or `!= NULL` ; must use `IS NULL` or `IS NOT NULL` .
3. Any arithmetic expression with NULL → **result is NULL**.
4. Any comparison with NULL → **result is UNKNOWN** (Three-Valued Logic).
5. `COALESCE(col, default)` replaces NULL with a specified default value.
6. Aggregate functions (SUM, AVG, MAX, MIN) **ignore NULLs** automatically.
7. `COUNT(*)` counts all rows; `COUNT(column)` excludes NULL rows.

⚡ Quick Recall

NULL = Unknown/Missing → ≠ Zero/Empty String → `IS NULL` / `IS NOT NULL` → Arithmetic→NULL,
Comparison→UNKNOWN → `COALESCE (default)` → Aggregates ignore NULLs (except `COUNT*`)